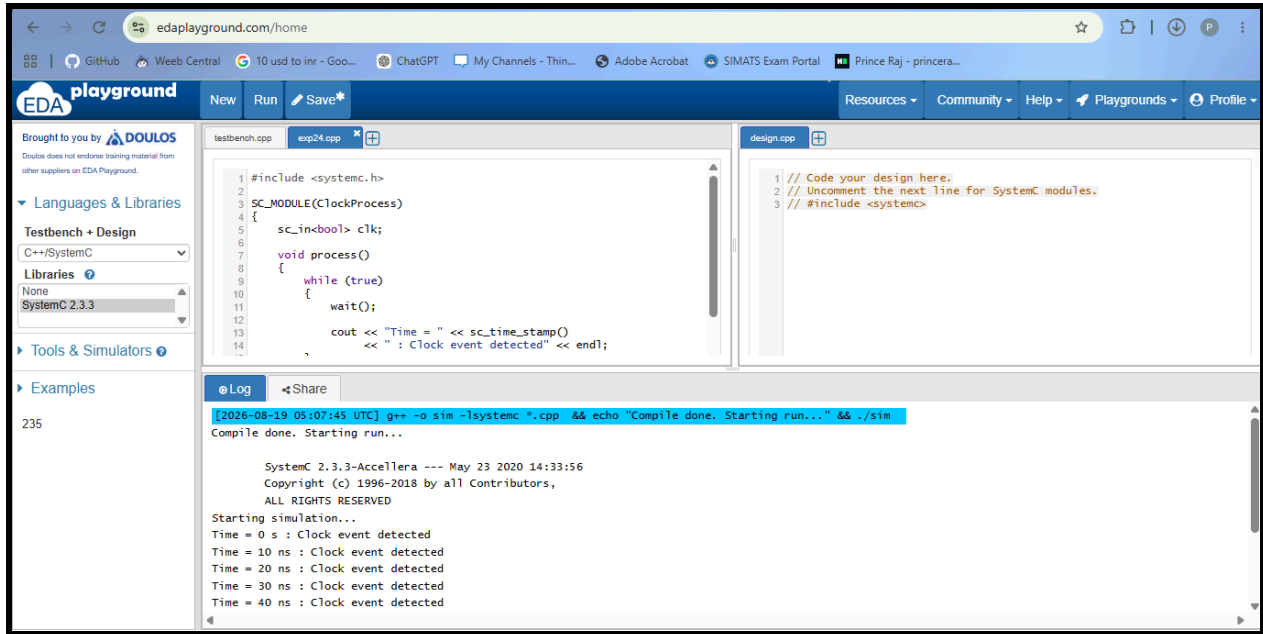




C++ Program:

```
#include <systemc.h>
```

```
SC_MODULE(ClockProcess)
```

```
{
```

```
    sc_in<bool> clk;
```

```
    void process()
```

```
    {
```

```
        while (true)
```

```
        {
```

```
            wait();
```

```
            cout << "Time = " << sc_time_stamp()
```

```
                << " : Clock event detected" << endl;
```

```
        }
```

```
    }
```

```
SC_CTOR(ClockProcess)
```

```
{
```

```
    SC_THREAD(process);
```

```
    sensitive << clk.pos();
```

```
}
```

```
int sc_main(int argc, char* argv[])
{
    sc_clock clk("clk", 10, SC_NS);

    ClockProcess cp("ClockProcess");
    cp.clk(clk);

    cout << "Starting simulation..." << endl;

    sc_start(100, SC_NS);

    cout << "Simulation completed." << endl;

    return 0;
}
```